Samples and Populations

Collecting Samples to Make Inferences About Large Populations

Colin Hannigan

CSB/SJU

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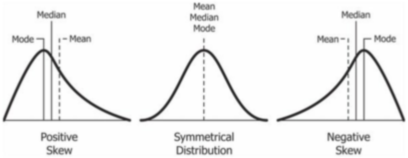
WDWTALT

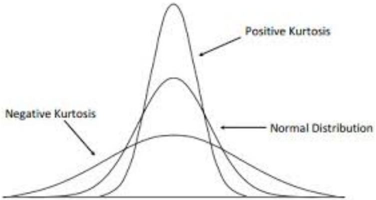
Measures of Central Tendency:

- Mean (arithmetic average)
- Median (middle-est value)
- Mode (most common value)

Measures of CT Distribution:

- Skewness (left or right shift)
- Kurtosis (up or down shift)
- Normality (standard)





Population is the full collection of things we want to study.

Sample is the subset of the population that we actually examine.

Two important things to make clear when collecting a sample:

1. The method of selection (how did you choose?)
2. The number of units (how many cases are in the sample?)

Questions to Discuss

1. Why do we sample?
2. Is sampling “scientific”?
3. Do we have to sample in research?

Types of Samples

Probability: each element has a known probability of being sampled

Nonprobability: the probability of selecting an element is unknown

Random: each element has an equal chance of being selected

Systematic: elements are chosen systematically (e.g., every 9th element)

Stratified: elements are selected from each strata (i.e., group) according to their proportions in the population

Cluster: clusters (i.e., groups of samples) are selected

Some Examples of Sampling Bias

Undercoverage bias: (i.e., exclusion bias) Sampling modes sometimes miss certain demographics or subsets of the population (e.g., phone surveys only reach people with phones).

Self-selection bias: (i.e., voluntary response bias) Participants who are more willing to take part in a study are not always reflective of the population (e.g., people self-select into taking surveys)

Survivorship bias: Some selection criteria can favor and over-cover the people who meet them (e.g., surveying only current students misses those who didn't apply, chose a different school, or transferred out)

Non-response bias: Samples miss people who don't participate, and their perspectives or demographics may constitute a systematic bias.

Recall bias: Some participants may not remember, mis-remember, satisfice or try to appear more "socially desirable" with the information they share.

Observer bias: The researcher can influence the participants unintentionally (e.g., agreement, enthusiasm, acknowledgement, tone, etc. may prime participants).

How do we avoid bias in sampling?